

1 Matrix Elements

Define the following matrix elements to make things go easier.

$$\tau_{ij}^{ab} = t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a \quad (1)$$

$$\tilde{\tau}_{ij}^{ab} = \frac{1}{2} t_{ij}^{ab} + t_i^a t_j^b \quad (2)$$

1.1 One Particle

$$\bar{H}_1 = \sum_{pq} F_{pq} N [\hat{a}_p^\dagger \hat{a}_q] \quad (3)$$

$$F_{ij} = f_{ij} + \sum_a t_j^a f_{ia} + \sum_{ka} t_k^a \langle ki || aj \rangle + \frac{1}{2} \sum_{kab} \tau_{kj}^{ba} \langle ki || ba \rangle \quad (4)$$

$$F_{ab} = f_{ab} - \sum_i t_i^a f_{ib} + \sum_{ic} t_i^c \langle ia || cb \rangle - \frac{1}{2} \sum_{ijc} \tau_{ij}^{ca} \langle ij || cb \rangle \quad (5)$$

$$F_{ia} = f_{ia} - \sum_{jb} t_j^b \langle ij || ab \rangle \quad (6)$$

Note that F_{ai} is the expression for the T_1 residual. At convergence of CCSD, this residual is zero.

$$F_{ai} = 0 \quad (7)$$

1.2 Two Particle

$$\bar{H}_2 = \sum_{pqrs} W_{pqrs} N [\hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r] \quad (8)$$

$$W_{ijkl} = \langle ij || kl \rangle + P_{kl}^- \sum_a t_k^a \langle ij || al \rangle + \frac{1}{2} \sum_{ab} \tau_{kl}^{ab} \langle ij || ab \rangle \quad (9)$$

$$W_{iajb} = \langle ia || jb \rangle - \sum_k t_k^a \langle ik || jb \rangle + \sum_c t_j^c \langle ia || cb \rangle - \sum_{kc} (t_{jk}^{ca} + t_j^c t_k^a) \langle ik || cb \rangle \quad (10)$$

$$W_{abcd} = \langle ab || cd \rangle - P_{ab}^- \sum_k t_k^a \langle kb || cd \rangle + \frac{1}{2} \sum_{kl} \tau_{kl}^{ab} \langle kl || cd \rangle \quad (11)$$

$$\begin{aligned} W_{iajk} &= \langle ia || jk \rangle + \sum_b t_{jk}^{ba} f_{ib} - \sum_l t_l^a \langle li || jk \rangle + P_{jk}^- \sum_b t_k^b \langle ia || jb \rangle - P_{jk}^- \sum_{lb} t_k^b t_l^a \langle li || bj \rangle \\ &+ \frac{1}{2} P_{jk}^- \sum_{bc} t_j^b t_k^c \langle ia || bc \rangle - \frac{1}{2} P_{jk}^- \sum_{lbc} t_j^b t_k^c t_l^a \langle il || bc \rangle + P_{jk}^- \sum_{lb} t_{kl}^{ab} \langle li || bj \rangle + \frac{1}{2} \sum_{bc} t_{jk}^{bc} \langle ia || bc \rangle + P_{jk}^- \sum_{lbc} t_{lk}^{ca} t_j^b \langle il || bc \rangle \\ &- \frac{1}{2} \sum_{lbc} t_{jk}^{bc} t_l^a \langle li || bc \rangle + \sum_{lbc} t_{jk}^{ac} t_l^b \langle li || bc \rangle \end{aligned} \quad (12)$$

Simplify.

$$\begin{aligned} W_{iajk} &= \langle ia || jk \rangle + \sum_b t_{jk}^{ba} F_{ib} - \sum_l t_l^a W_{lij} - P_{jk}^- \sum_c t_j^c \left(\langle ia || kc \rangle - \sum_{lb} t_{kl}^{ab} \langle li || bc \rangle \right) \\ &+ \frac{1}{2} \sum_{bc} \tau_{jk}^{bc} \langle ia || bc \rangle + P_{jk}^- \sum_{lb} t_{kl}^{ab} \langle li || bj \rangle \end{aligned} \quad (13)$$

$$\begin{aligned}
W_{ab|ic} = & \langle ab||ic \rangle - \sum_m t_{mi}^{ab} f_{mc} + \sum_e t_i^e \langle ab||ec \rangle - P_{ab}^- \sum_m t_m^a \langle mb||ic \rangle - P_{ab}^- \sum_{me} t_m^b t_i^e \langle ma||ce \rangle \\
& + \frac{1}{2} P_{ab}^- \sum_{mn} t_m^a t_n^b \langle mn||ci \rangle + \frac{1}{2} P_{ab}^- \sum_{me} t_m^b t_i^e t_n^a \langle mn||ce \rangle + \frac{1}{2} \sum_{mn} t_{mn}^{ab} \langle mn||ci \rangle - P_{ab}^- \sum_{me} t_{im}^{ae} \langle mb||ec \rangle \\
& + \frac{1}{2} \sum_{mne} t_{mn}^{ab} t_i^e \langle mn||ec \rangle + \sum_{mne} t_{ni}^{ba} t_m^e \langle mn||ec \rangle + P_{ab}^- \sum_{mne} t_{ni}^{ea} t_m^b \langle mn||ec \rangle \quad (14)
\end{aligned}$$

Simplify.

$$\begin{aligned}
W_{ab|ic} = & \langle ab||ic \rangle - \sum_m t_{mi}^{ab} F_{mc} + \sum_e t_i^e W_{abec} - P_{ab}^- \sum_m t_m^a \left(\langle mb||ic \rangle + \sum_{ne} t_{ni}^{eb} \langle mn||ec \rangle \right) \\
& + \frac{1}{2} \sum_{mn} \tau_{mn}^{ab} \langle mn||ci \rangle - P_{ab}^- \sum_{me} t_{im}^{ae} \langle mb||ec \rangle \quad (15)
\end{aligned}$$

$$W_{ijak} = \langle ij||ak \rangle + \sum_e t_k^e \langle ij||ae \rangle \quad (16)$$

$$W_{iabc} = \langle ia||bc \rangle - \sum_m t_m^a \langle im||bc \rangle \quad (17)$$

The value W_{abij} is simply the T_2 residual, which at CCSD convergence should be zero.

$$W_{abij} = 0 \quad (18)$$

$$W_{ijab} = \langle ij||ab \rangle \quad (19)$$

2 Right Vector

Now that we have the diagrams, we need to compute the following.

$$\langle \Phi_\mu | \bar{H} \hat{R} | \Phi_0 \rangle \quad (20)$$

2.1 Zero-order terms

$$\langle \Phi_0 | \bar{H} \hat{R}_0 | \Phi_0 \rangle = r_0 E_{CCSD} \quad (21)$$

$$\langle \Phi_i^a | \bar{H} \hat{R}_0 | \Phi_0 \rangle = 0 \quad (22)$$

$$\langle \Phi_{ij}^{ab} | \bar{H} \hat{R}_0 | \Phi_0 \rangle = 0 \quad (23)$$

2.2 First-order terms

$$\langle \Phi_0 | \bar{H} \hat{R}_1 | \Phi_0 \rangle = \sum_{me} r_m^e F_{me} \quad (24)$$

$$\langle \Phi_i^a | \bar{H} \hat{R}_1 | \Phi_0 \rangle = \sum_m r_m^a F_{mi} - \sum_e r_i^e F_{ae} - \sum_{me} r_m^e W_{maei} \quad (25)$$

$$\langle \Phi_{ij}^{ab} | \bar{H} \hat{R}_1 | \Phi_0 \rangle = P_{ij}^- \sum_e r_i^e W_{abej} - P_{ab}^- \sum_m r_m^a W_{mbij} \quad (26)$$

2.3 Second-order terms

$$\left\langle \Phi_0 \left| \bar{H} \hat{R}_2 \right| \Phi_0 \right\rangle = \frac{1}{4} \sum_{mnef} r_{mn}^{ef} W_{mnef} \quad (27)$$

$$\left\langle \Phi_i^a \left| \bar{H} \hat{R}_2 \right| \Phi_0 \right\rangle = - \sum_{me} r_{mi}^{ae} F_{me} + \frac{1}{2} \sum_{mne} r_{mn}^{ea} W_{mnei} - \frac{1}{2} \sum_{mef} r_{mi}^{ef} W_{maef} \quad (28)$$

$$\left\langle \Phi_{ij}^{ab} \left| \bar{H} \hat{R}_2 \right| \Phi_0 \right\rangle = -P_{ij}^- \sum_m r_{mj}^{ab} F_{mi} + P_{ab}^- \sum_e r_{ij}^{eb} F_{ae} + \frac{1}{2} \sum_{mn} r_{mn}^{ab} W_{mnij} + \frac{1}{2} \sum_{ef} r_{ij}^{ef} W_{abef} - P_{ij}^- P_{ab}^- \sum_{me} r_{im}^{ae} W_{mbej} \quad (29)$$

2.4 Combined expressions

$$\left\langle \Phi_0 \left| \bar{H} \hat{R} \right| \Phi_0 \right\rangle = r_0 E_{CCSD} + \sum_{me} r_m^e F_{me} + \frac{1}{4} \sum_{mnef} r_{mn}^{ef} W_{mnef} \quad (30)$$

$$\left\langle \Phi_i^a \left| \bar{H} \hat{R} \right| \Phi_0 \right\rangle = \sum_m r_m^a F_{mi} - \sum_e r_i^e F_{ae} - \sum_{me} r_m^e W_{maei} - \sum_{me} r_{mi}^{ae} F_{me} + \frac{1}{2} \sum_{mne} r_{mn}^{ae} W_{mnei} - \frac{1}{2} \sum_{mef} r_{mi}^{ef} W_{maef} \quad (31)$$

$$\begin{aligned} \left\langle \Phi_{ij}^{ab} \left| \bar{H} \hat{R} \right| \Phi_0 \right\rangle &= P_{ij}^- \sum_e r_i^e W_{abej} - P_{ab}^- \sum_m r_m^a W_{mbij} - P_{ij}^- \sum_m r_{mj}^{ab} F_{mi} + P_{ab}^- \sum_e r_{ij}^{eb} F_{ae} + \frac{1}{2} \sum_{mn} r_{mn}^{ab} W_{mnij} \\ &\quad + \frac{1}{2} \sum_{ef} r_{ij}^{ef} W_{abef} - P_{ij}^- P_{ab}^- \sum_{me} r_{im}^{ae} W_{mbej} \end{aligned} \quad (32)$$